

The Rules of Harmony

*A musician's field guide to scales, chords, harmony,
counterpoint and rhythm*

50 scales · 37 chord types · the rules that connect them

Generated from the rule engine of *midi-composer-mcp*. Every table, example and catalogue entry in this book is produced by the same deterministic code that the composing tools run.

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How to read this book

This book explains the building blocks of Western tonal music — notes, intervals, scales, chords, keys — and the rules composers use to connect them: harmony, voice leading, reharmonization, melody craft, counterpoint, rhythm and form. It starts from zero, but it does not stay there: each chapter ends where a working musician needs it to.

Some conventions used throughout:

- **Note names** are letters with accidentals: C, F#, Bb. A name without an octave number is a *pitch class* — every C on the piano. With a number it is one concrete key: C4 is middle C, C5 the C above it.
- **Keyboard diagrams** show one or two octaves from C. The home note (root) is dark, the other members are orange, and each marked key is labelled with its correct spelling.
- **Degrees** are written 1, 2, b3, #4... — the scale step, altered relative to the major scale. **Roman numerals** name chords by the degree they stand on: I, ii, V7, bVII.
- **Pro notes** (tinted boxes) add detail for experienced readers. Beginners can skip them without losing the thread.

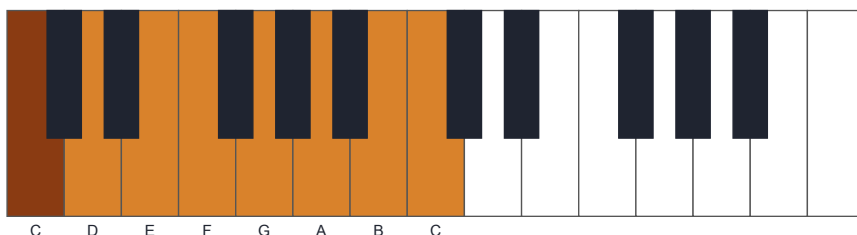
Why spelling matters. Bb and A# are the same key on a piano, but not the same note in music. F major contains Bb, never A#, because a scale uses each letter once. Correct spelling keeps intervals readable (a third always spans three letters) and is why this book, like the tools behind it, writes Cb, E#, or a double flat when the theory calls for one.

CHAPTER 1

Notes, octaves and spelling

Twelve pitches repeat in every octave. Seven letters name them, and accidentals fill the gaps.

The piano keyboard makes the system visible: seven white keys named C D E F G A B, and five black keys between them. Moving from any key to its nearest neighbour is a **semitone** (half step); two semitones make a **whole tone**. Between E–F and B–C there is no black key, so those white keys are only a semitone apart.



A **sharp** (#) raises a note by a semitone and a **flat** (b) lowers it; a double sharp or double flat moves it two. So every black key has two names — C# is also Db — and even white keys can be respelled: E# sounds as F, Cb as B. These are **enharmonic** spellings: same sound, different meaning.

An **octave** is the distance to the next note with the same letter; its frequency is exactly double. Octaves are numbered from C upward, so B3 is directly below C4. That is why Cb4 is the same key as B3, and B#3 the same key as C4 — the number belongs to the letter, not to the key.

Concrete pitches, MIDI and frequency

Note	MIDI number	Frequency (Hz)
A0	21	27.50
C2	36	65.41
C3	48	130.81
C4 (middle C)	60	261.63
A4 (tuning A)	69	440.00
C5	72	523.25
C6	84	1046.50
C8	108	4186.01

Pro note. MIDI numbers run from 0 (C-1) to 127 (G9); middle C is 60 and each semitone adds one. In equal temperament every semitone multiplies the frequency by the twelfth root of two (about 1.0595). Other temperaments — just intonation, the gamelan tunings behind pelog — divide the octave differently; this book stays in twelve equal steps.

CHAPTER 2

Intervals

An interval is the distance between two notes, named by how many letters it spans and how many semitones it holds.

Count the letters from the lower note to the upper one, including both: C up to E spans C-D-E, three letters, so it is a **third**. The number alone is not enough — C–E and C–E $\flat$ are both thirds — so a **quality** is added: **major** or **minor** for 2nds, 3rds, 6ths and 7ths; **perfect** for unisons, 4ths, 5ths and octaves. Stretch any of them by a semitone and it becomes **augmented**; shrink it and it becomes **diminished**.

Interval	Short	Semitones	From C	Character
perfect unison	P1	0	C → C	the same note
minor second	m2	1	C → D $\flat$	sharp rub, tense
major second	M2	2	C → D	a step; mild
minor third	m3	3	C → E $\flat$	dark, sad colour
major third	M3	4	C → E	bright, stable
perfect fourth	P4	5	C → F	open; wants to fall to the 3rd
augmented fourth	A4	6	C → F $\sharp$	the tritone: restless
diminished fifth	d5	6	C → G $\flat$	the tritone, spelled as a 5th
perfect fifth	P5	7	C → G	hollow, strong
minor sixth	m6	8	C → A $\flat$	sweet, melancholy
major sixth	M6	9	C → A	warm, open
minor seventh	m7	10	C → B $\flat$	bluesy tension
major seventh	M7	11	C → B	yearning, bright
perfect octave	P8	12	C → C	the octave: the same note higher

Same sound, different interval

Because intervals are counted by letters, C–F $\sharp$ is an augmented 4th but C–G $\flat$ a diminished 5th, although both are six semitones. C up to B $\sharp$ is an augmented 7th (twelve semitones!), and B $\sharp$ 3 up to C4 a diminished second of zero semitones. The spelling tells you how the note behaves: an augmented 4th wants to open outwards, a diminished 5th to close inwards.

Consonance and dissonance

In two-voice writing the **consonances** are the perfect unison, octave and fifth (the *perfect consonances*) and the major and minor thirds and sixths (the *imperfect consonances*). Everything else — seconds, sevenths, the tritone, and, when two voices sound alone, the perfect fourth — is a **dissonance** that needs careful handling. Chapter 9 turns this into rules.

Compound intervals and inversion

Intervals wider than an octave are **compound**: a 9th is an octave plus a 2nd, a 10th an octave plus a 3rd (C₄ → E₅ is a major tenth). Turning an interval upside down (moving the lower note up an octave) **inverts** it: the numbers add to nine and the qualities swap — major↔minor, augmented↔diminished, perfect stays perfect. So a major 3rd inverts to a minor 6th.

CHAPTER 3

Scales

A scale is a set of notes ordered from a home note. This chapter explains how scales are built, then catalogues all 50 of them.

A **scale** picks some of the twelve pitch classes and orders them from a **tonic**. Its identity is its pattern of steps: the major scale is whole–whole–half–whole–whole–whole–half (W W H W W W H). Start that pattern on any note and you get that note's major scale; the letters each appear once, which is why the spelling comes out right (F major has B $\flat$, D major has F $\sharp$ and C $\sharp$).

Scales are described by their **degrees** relative to the major scale: natural minor is 1 2 $\flat$ 3 4 5 $\flat$ 6 $\flat$ 7, dorian 1 2 $\flat$ 3 4 5 6 $\flat$ 7. The degree list is the fastest way to compare scales — and to transpose them: to play any scale on any root, apply its degrees to that root's major scale.

Modes are scales that share their notes with a parent but start elsewhere. D dorian uses the white keys from D; so do E phrygian and G mixolydian. Same notes, different home — and a different mood, because the half steps fall in new places relative to the tonic.

Try it. Play C major (all white keys from C), then A natural minor (all white keys from A). The notes are the same; only the home note changed. That pair — a major key and the minor key three semitones below — are *relative* keys.

The major scale and its modes

Seven scales that share one set of notes, each starting on a different degree. Brightest to darkest: lydian, major (ionian), mixolydian, dorian, natural minor (aeolian), phrygian, locrian.

Major *also ionian, maj*

Degrees 1 2 3 4 5 6 7

On C C D E F G A B

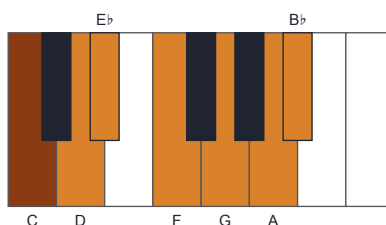
The standard major scale (do-re-mi); bright, stable, and the bedrock of Western tonal music.

**Dorian**

Degrees 1 2 $\flat$ 3 4 5 6 $\flat$ 7

On C C D E $\flat$ F G A B $\flat$

A minor mode with a bright raised 6th; the hopeful, jazzy minor of folk and modal jazz.

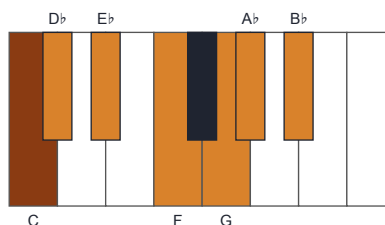


Phrygian

Degrees 1 $\flat 2$ $\flat 3$ 4 5 $\flat 6$ $\flat 7$

On C C C D $\flat$ E $\flat$ F G A $\flat$ B $\flat$

A minor mode with a flat 2nd; dark and tense, with a Spanish/flamenco undercurrent.

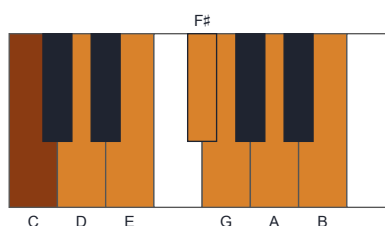


Lydian

Degrees 1 2 3 #4 5 6 7

On C C C D E F# G A B

A major mode with a raised 4th; dreamy, floating and luminous, a film-score favorite.

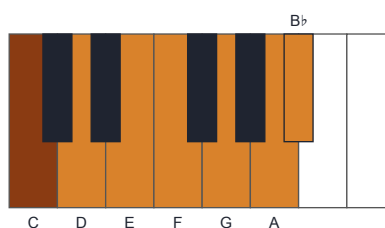


Mixolydian *also dominant scale*

Degrees 1 2 3 4 5 6 $\flat 7$

On C C C D E F G A B $\flat$

A major mode with a flat 7th; the dominant-7th sound of blues, rock, funk and Celtic music.

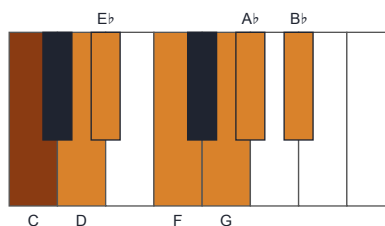


Natural Minor *also minor, min, aeolian*

Degrees 1 2 $\flat 3$ 4 5 $\flat 6$ $\flat 7$

On C C C D E $\flat$ F G A $\flat$ B $\flat$

The standard minor scale and relative minor of the major scale; wistful and melancholic.

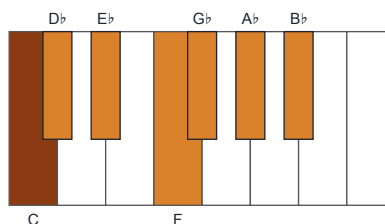


Locrian

Degrees 1 $\flat 2$ $\flat 3$ 4 $\flat 5$ $\flat 6$ $\flat 7$

On C C C D $\flat$ E $\flat$ F G $\flat$ A $\flat$ B $\flat$

A diminished mode with flat 2nd and flat 5th; unstable and rarely a tonic, home of the m7 $\flat 5$ chord.



Minor variants

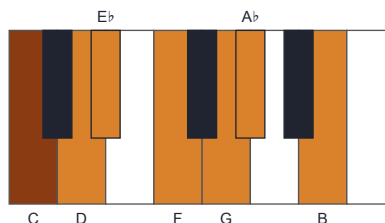
Natural minor has no leading tone (the note a semitone below the tonic). Raising the 7th gives harmonic minor; raising the 6th as well smooths the resulting augmented second and gives melodic minor.

Harmonic Minor

Degrees 1 2 $\flat$ 3 4 5 $\flat$ 6 7

On C C C D E $\flat$ F G A $\flat$ B

Natural minor with a raised 7th, adding a strong leading tone and a dramatic augmented-2nd step.



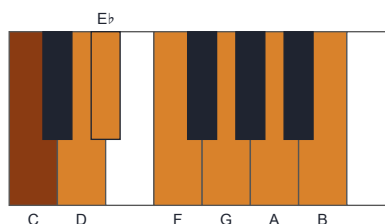
Melodic Minor *also jazz minor, melodic minor*

ascending

Degrees 1 2 $\flat$ 3 4 5 6 7

On C C C D E $\flat$ F G A B

Natural minor with raised 6th and 7th; smooths melodies to the tonic and is the parent scale of much jazz.



Modes of melodic minor

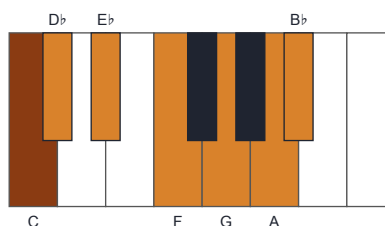
The modern jazz palette: each mode fits a particular chord colour, from the lydian-dominant 7#11 to the altered dominant.

Dorian $\flat$ 2 *also phrygian #6, assyrian*

Degrees 1 $\flat$ 2 $\flat$ 3 4 5 6 $\flat$ 7

On C C C D $\flat$ E $\flat$ F G A B $\flat$

The 2nd mode of melodic minor; dorian brightness over a phrygian flat-2nd, restless and exotic.

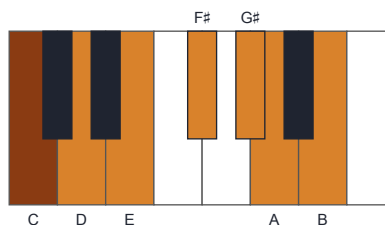


Lydian Augmented

Degrees 1 2 3 #4 #5 6 7

On C C C D E F# G# A B

The 3rd mode of melodic minor; a lydian scale with a raised 5th, shimmering and unresolved.

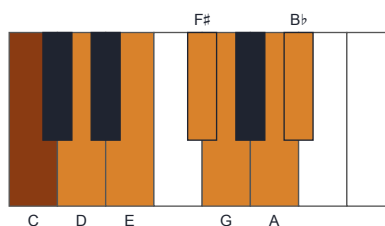


Lydian Dominant *also lydian $\flat$ 7, mixolydian #4, overtone, acoustic*

Degrees 1 2 3 #4 5 6 $\flat$ 7

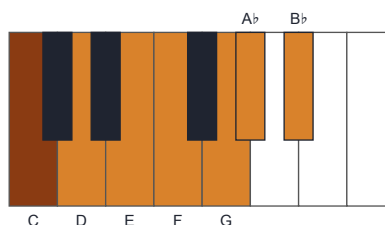
On C C C D E F# G A B $\flat$

The 4th mode of melodic minor: a dominant scale with a raised 4th, matching the natural overtone series.

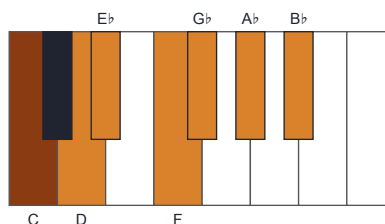


Mixolydian $\flat 6$ *also melodic major, hindu*Degrees 1 2 3 4 5 $\flat 6$ $\flat 7$ On C C D E F G $A\flat$ $B\flat$

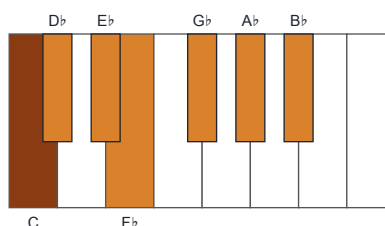
Mixolydian with a flat 6th; a dominant scale that leans melancholy, good moving to minor.

**Locrian $\sharp 2$** *also half-diminished scale, aeolian $\flat 5$* Degrees 1 2 $\flat 3$ 4 $\flat 5$ $\flat 6$ $\flat 7$ On C C D $E\flat$ F $G\flat$ $A\flat$ $B\flat$

The 6th mode of melodic minor; locrian with a natural 2nd, the smoothest scale for $m7\flat 5$ chords.

**Altered** *also super locrian, altered dominant, diminished whole tone*Degrees 1 $\flat 2$ $\flat 3$ $\flat 4$ $\flat 5$ $\flat 6$ $\flat 7$ On C C D $\flat$ $E\flat$ F $\flat$ $G\flat$ $A\flat$ $B\flat$

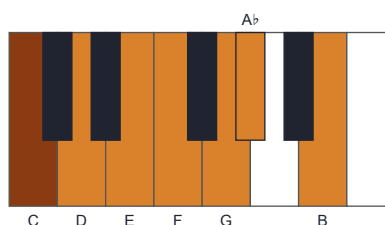
The 7th mode of melodic minor; packed with every altered tension ($\flat 9$ $\sharp 9$ $\sharp 11$ $\flat 13$) for maximally tense dominants.

**Harmonic major, harmonic minor relatives and folk colours**

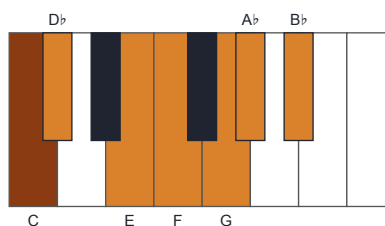
Scales with augmented seconds and other distinctive steps — the sound of flamenco, klezmer, Eastern Europe and the Middle East.

Harmonic MajorDegrees 1 2 3 4 5 $\flat 6$ 7On C C D E F G $A\flat$ B

Major with a flat 6th; blends major brightness with a touch of minor shadow and an augmented-2nd step.

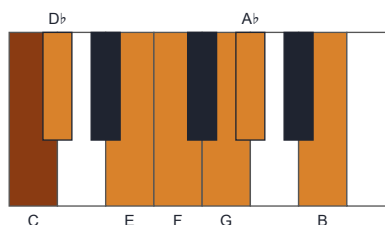
**Phrygian Dominant** *also spanish phrygian, freygish, ahava raba*Degrees 1 $\flat 2$ 3 4 5 $\flat 6$ $\flat 7$ On C C D $\flat$ E F G $A\flat$ $B\flat$

The 5th mode of harmonic minor; a major 3rd over a phrygian $\flat 2$, the sound of flamenco and Klezmer.

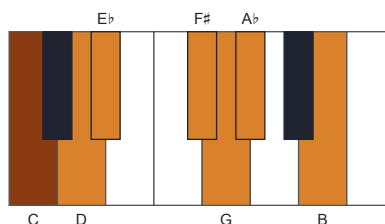


Double Harmonic *also byzantine, arabic, gypsy major*Degrees 1 $\flat$ 2 3 4 5 $\flat$ 6 7On C C C D $\flat$ E F G A $\flat$ B

Two augmented 2nds give this scale a vivid Middle-Eastern/Byzantine character.

**Hungarian Minor** *also gypsy minor*Degrees 1 2 $\flat$ 3 #4 5 $\flat$ 6 7On C C C D E $\flat$ F# G A $\flat$ B

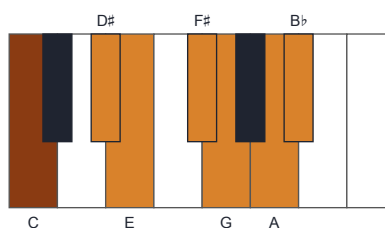
Harmonic minor with a raised 4th; the gypsy-minor scale with two dramatic augmented 2nds.

**Hungarian Major**

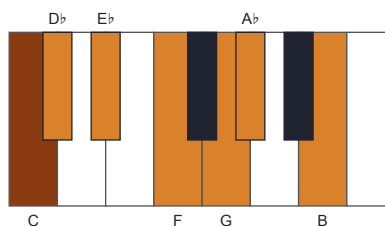
Degrees 1 #2 3 #4 5 6 7

On C C C D# E F# G A B $\flat$

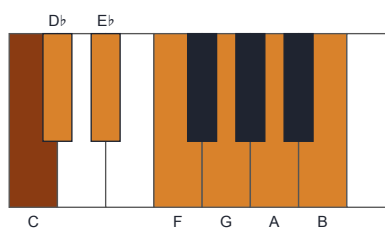
A bright, fiery scale opening on a raised 2nd and carrying a flat 7th; rooted in Hungarian folk music.

**Neapolitan Minor**Degrees 1 $\flat$ 2 $\flat$ 3 4 5 $\flat$ 6 7On C C C D $\flat$ E $\flat$ F G A $\flat$ B

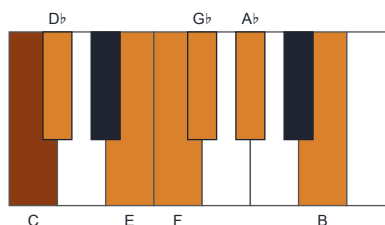
Harmonic minor with a flat 2nd; somber and exotic, resolving with a strong leading tone.

**Neapolitan Major**Degrees 1 $\flat$ 2 $\flat$ 3 4 5 6 7On C C C D $\flat$ E $\flat$ F G A B

Melodic minor with a flat 2nd; smooth and slightly unusual, major on top of a phrygian start.

**Persian**Degrees 1 $\flat$ 2 3 4 $\flat$ 5 $\flat$ 6 7On C C C D $\flat$ E F G $\flat$ A $\flat$ B

A flat-2nd, flat-5th scale with two augmented 2nds; intensely ornamental and Middle-Eastern.

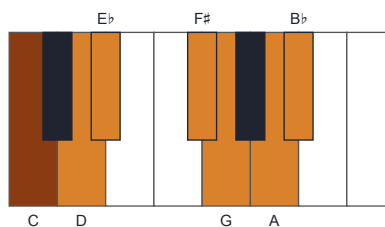


Ukrainian Dorian *also romanian minor, altered dorian*

Degrees 1 2 $\flat$ 3 #4 5 6 $\flat$ 7

On C C D E $\flat$ F# G A B $\flat$

Dorian with a raised 4th; plaintive and folk-like across Eastern European and Klezmer music.

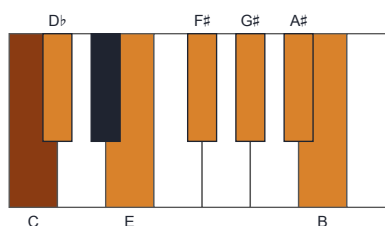


Enigmatic

Degrees 1 $\flat$ 2 3 #4 #5 #6 7

On C C D $\flat$ E F# G# A# B

Verdi's invented scale mixing whole tones with leading tones; strange, bright and unresolved.



Pentatonics and blues

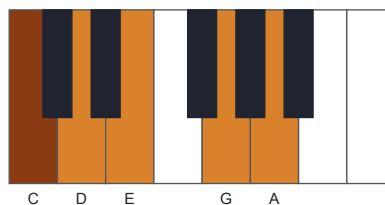
Five- and six-note scales without the most tense intervals; almost any combination of their notes sounds good.

Major Pentatonic *also pentatonic, pentatonic major*

Degrees 1 2 3 5 6

On C C D E G A

A five-note major scale with no semitones; open, singable and consonant in any combination.

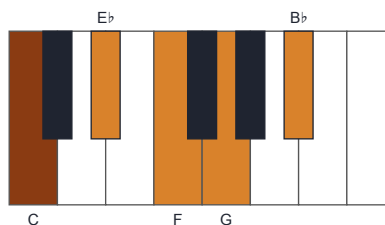


Minor Pentatonic *also pentatonic minor*

Degrees 1 $\flat$ 3 4 5 $\flat$ 7

On C C E $\flat$ F G B $\flat$

The five-note minor scale at the core of blues, rock and pop soloing.

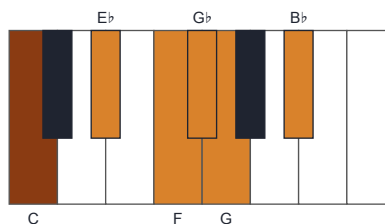


Blues *also minor blues, blues minor*

Degrees 1 $\flat$ 3 4 $\flat$ 5 5 $\flat$ 7

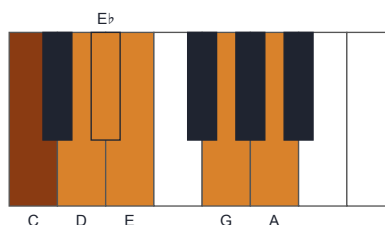
On C C E $\flat$ F G $\flat$ G B $\flat$

Minor pentatonic plus a chromatic 'blue note' ($\flat$ 5); the gritty, vocal sound of the blues.



Major Blues *also blues major*Degrees 1 2 $\flat 3$ 3 5 6On C C C D $E\flat$ E G A

Major pentatonic with an added $\flat 3$ blue note; bright yet bluesy, common in country and rock.

**Egyptian** *also suspended pentatonic*Degrees 1 2 4 5 $\flat 7$ On C C D F G B $\flat$

A suspended-sounding mode of the major pentatonic built on stacked 4ths and 5ths; open and unresolved.

**Symmetric scales**

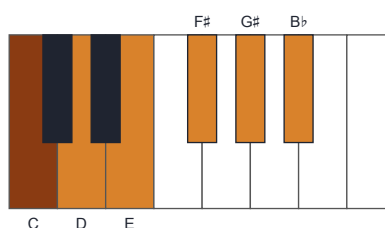
Scales built from a repeating pattern of steps. Because they repeat, they have fewer distinct transpositions and blur the sense of a home note.

Whole Tone *also wholetone, messiaen mode 1, messiaen*

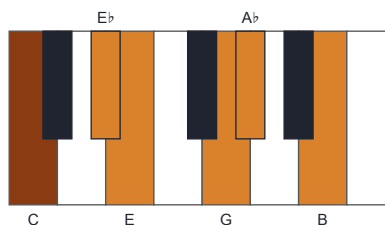
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Degrees 1 2 3 $\sharp 4$ $\sharp 5$ $\flat 7$ On C C D E F $\sharp$ G $\sharp$ B $\flat$

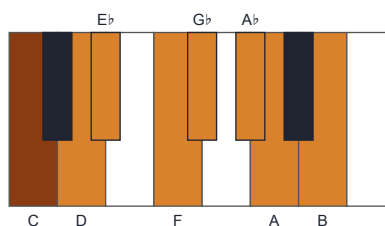
Six notes a whole step apart; dreamlike and rootless with no leading tone, beloved by Debussy.

**Augmented** *also symmetric augmented*Degrees 1 $\flat 3$ 3 5 $\flat 6$ 7On C C $E\flat$ E G A $\flat$ B

A six-note symmetric scale alternating minor 3rds and half steps; stark and built from two augmented triads.

**Diminished Whole-half** *also diminished, octatonic, whole-half diminished*Degrees 1 2 $\flat 3$ 4 $\flat 5$ $\flat 6$ 6 7On C C D $E\flat$ F G $\flat$ A $\flat$ A B

An eight-note symmetric scale alternating whole and half steps; played over diminished-7th chords.

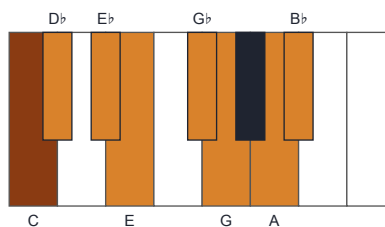


Diminished Half-whole *also dominant diminished, half-whole diminished, messiaen mode 2, messiaen 2*

Degrees 1 $\flat 2 \flat 3 3 \flat 5 5 6 \flat 7$

On C C D $\flat$ E $\flat$ E G $\flat$ G A B $\flat$

The octatonic scale starting with a half step; the go-to choice over dominant 7 $\flat 9$ /#9 chords.

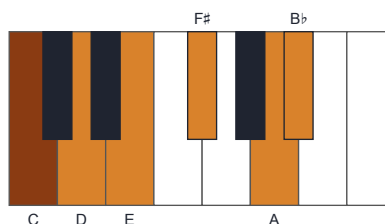


Prometheus *also mystic*

Degrees 1 2 3 #4 6 $\flat 7$

On C C D E F# A B $\flat$

Scriabin's six-note 'mystic' scale built from the mystic chord; hovering, ambiguous and tense.



Messiaen's modes of limited transposition

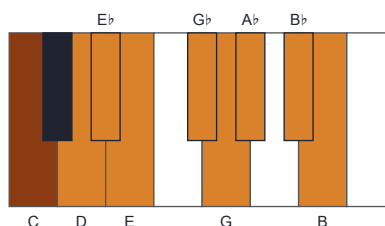
Olivier Messiaen's symmetric modes: each repeats inside the octave, so it has only a few transpositions before it returns to itself. Mode 1 is the whole-tone scale and mode 2 the half-whole octatonic scale.

Messiaen Mode 3 *also messiaen 3*

Degrees 1 2 $\flat 3 3 \flat 5 5 \flat 6 \flat 7 7$

On C C D E $\flat$ E G $\flat$ G A $\flat$ B $\flat$ B

Messiaen's third mode of limited transposition: nine notes, only four transpositions; luminous and chromatic.

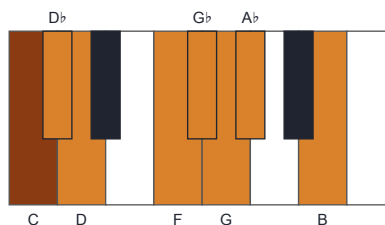


Messiaen Mode 4 *also messiaen 4*

Degrees 1 $\flat 2 2 4 \flat 5 5 \flat 6 7$

On C C D $\flat$ D F G $\flat$ G A $\flat$ B

Messiaen's fourth mode: two chromatic clusters a tritone apart; six transpositions, hushed and mysterious.

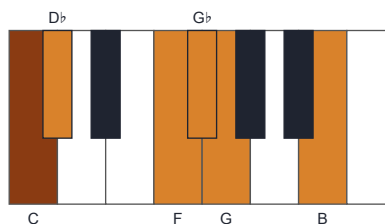


Messiaen Mode 5 *also messiaen 5*

Degrees 1 $\flat 2 4 \flat 5 5 7$

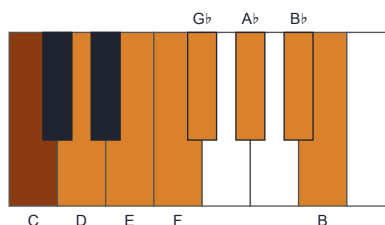
On C C D $\flat$ F G $\flat$ G B

Messiaen's fifth mode: a sparse six-note subset of mode 4; stark and bell-like.

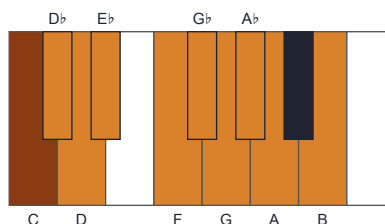


Messiaen Mode 6 *also messiaen 6*Degrees 1 2 3 4 $\flat 5$ $\flat 6$ $\flat 7$ 7On C C C D E F G $\flat$ A $\flat$ B $\flat$ B

Messiaen's sixth mode: whole tones with added semitones, repeating at the tritone; bright and unstable.

**Messiaen Mode 7** *also messiaen 7*Degrees 1 $\flat 2$ 2 $\flat 3$ 4 $\flat 5$ 5 $\flat 6$ 6 7On C C D $\flat$ D E $\flat$ F G $\flat$ G A $\flat$ A B

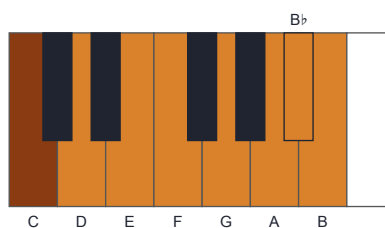
Messiaen's seventh mode: ten notes repeating at the tritone; dense, iridescent harmony.

**Bebop and eight-note scales**

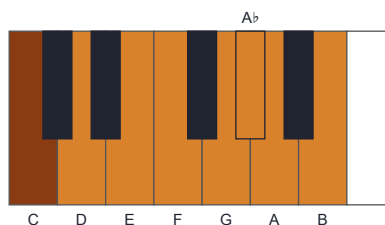
Seven-note scales with one extra passing note, so that in running eighth notes the chord tones fall on the beats.

Bebop Dominant *also bebop*Degrees 1 2 3 4 5 6 $\flat 7$ 7On C C C D E F G A B $\flat$ B

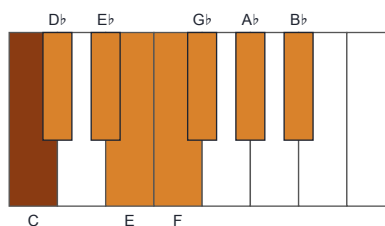
Mixolydian plus a passing major 7th; the extra note keeps chord tones on the beat in fast bebop lines.

**Bebop Major**Degrees 1 2 3 4 5 $\flat 6$ 6 7On C C C D E F G A $\flat$ A B

The major scale with a passing $\flat 6$ ($\sharp 5$); aligns chord tones to strong beats over major harmony.

**Spanish 8-tone** *also spanish gypsy, jewish 8-tone*Degrees 1 $\flat 2$ $\flat 3$ 3 4 $\flat 5$ $\flat 6$ $\flat 7$ On C C D $\flat$ E $\flat$ E F G $\flat$ A $\flat$ B $\flat$

An eight-note flamenco scale extending phrygian dominant with chromatic passing tones for both 3rds.



Japanese and world pentatonics

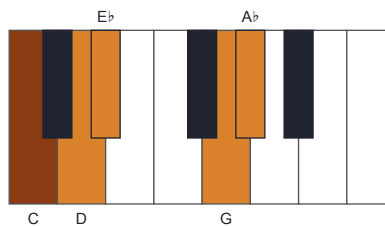
Five-note scales from Japanese and Indonesian traditions, here in 12-tone equal temperament.

Hirajoshi

Degrees 1 2 $\flat$ 3 5 $\flat$ 6

On C C D E $\flat$ G A $\flat$

A Japanese pentatonic with two semitone steps; sparse, dark and contemplative on koto and shakuhachi.

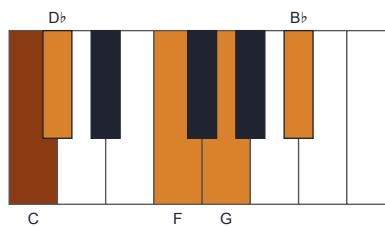


In Sen *also in, insen*

Degrees 1 $\flat$ 2 4 5 $\flat$ 7

On C C D $\flat$ F G B $\flat$

A dark Japanese pentatonic built on the flat 2nd; haunting and traditional.

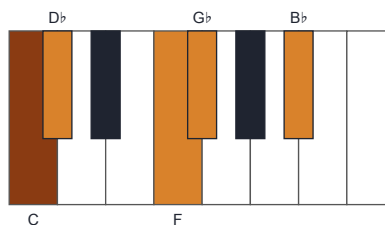


Iwato

Degrees 1 $\flat$ 2 4 $\flat$ 5 $\flat$ 7

On C C D $\flat$ F G $\flat$ B $\flat$

A somber Japanese pentatonic with two tritone-flavored tensions; tense and shadowy.

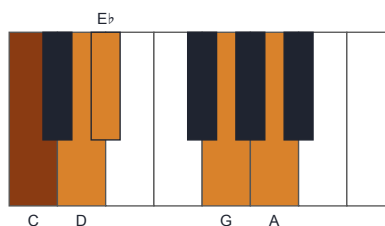


Kumoi *also kumojoshi*

Degrees 1 2 $\flat$ 3 5 6

On C C D E $\flat$ G A

A Japanese pentatonic with a bright opening and minor color; delicate and evocative.

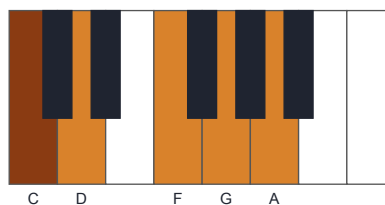


Yo

Degrees 1 2 4 5 6

On C C D F G A

A bright, semitone-free Japanese pentatonic used in folk songs and Buddhist chant.

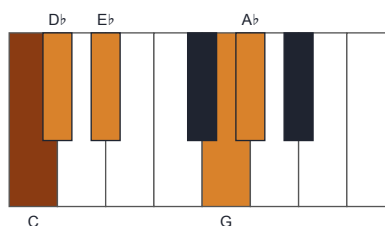


Balinese Pelog *also pelog*

Degrees 1 $\flat$ 2 $\flat$ 3 5 $\flat$ 6

On C C D $\flat$ E $\flat$ G A $\flat$

A five-note scale approximating Indonesian gamelan pelog tuning in 12-tone equal temperament; exotic and uneven.



The complete chromatic scale

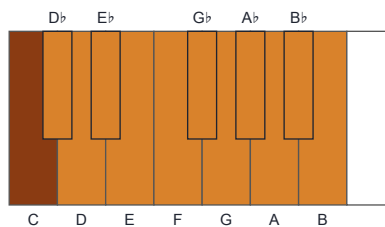
All twelve notes: not a key, but the full palette.

Chromatic

Degrees 1 $\flat$ 2 2 $\flat$ 3 3 4 $\flat$ 5 5 $\flat$ 6 6 $\flat$ 7 7

On C C C $\flat$ D D E $\flat$ E F G $\flat$ G A $\flat$ A B $\flat$ B

All twelve semitones; not a tonal scale but the complete palette for passing tones and atonal writing.



CHAPTER 4

Chords

Three or more notes sounding together. Most chords are stacks of thirds; this chapter shows how they are built and named, then lists all 37 types.

Take every other note of a scale — 1, 3, 5 — and you have a **triad**. Its quality depends on the thirds it is made of: major third + minor third gives a **major** triad (C E G), minor + major a **minor** triad (C E $\flat$ G), two minor thirds a **diminished** triad, two major thirds an **augmented** one. Keep stacking thirds and you get sevenths (1 3 5 7), ninths, elevenths and thirteenth.

A **chord symbol** is the root plus a suffix: C (major), Cm, C7, Cmaj7, Cm7 $\flat$ 5, C13. The suffix is case-sensitive in one place that matters: **m** is minor, **M** is major (CM7 = Cmaj7). A minus sign means minor in jazz charts (C-7 = Cm7), and in older charts a minus or plus before a number flattens or sharpens it (C7-9 = C7 $\flat$ 9, C7+5 = C7 $\sharp$ 5).

Inversions and slash chords

A chord is in **root position** when its root is the lowest note. With the 3rd in the bass it is in **first inversion**, with the 5th **second inversion**, with the 7th **third inversion**. A **slash chord** names the bass explicitly: C/E is C major over E (first inversion); C/D puts a non-chord tone in the bass. The inversion is named by which chord member is in the bass — the 5th in the bass is always second inversion, whatever the chord.

Pro note. Added-tone chords (add9, add4, 6) keep the triad and add a colour without the 7th; sus chords *replace* the 3rd with a 2nd or 4th. Upper extensions are usually voiced selectively: an 11th chord often drops its 3rd (which clashes with the 11th), a 13th often drops the 11th.

Triads and their relatives

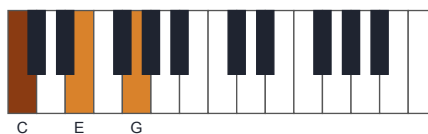
Major C

Formula 1 3 5

On C C E G

Also written Cmaj, CM

The bright, stable major triad: root, major 3rd, perfect 5th.



Minor Cm

Formula 1 $\flat$ 3 5

On C C E $\flat$ G

Also written Cmin, C-

The dark, sad minor triad: root, minor 3rd, perfect 5th.



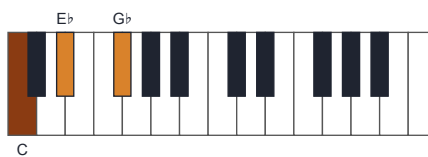
Diminished **Cdim**

Formula 1 $\flat$ 3 $\flat$ 5

On C C E $\flat$ G $\flat$

Also written C $^\circ$, Co

A tense triad of stacked minor 3rds (minor 3rd plus flat 5th) that wants to resolve.



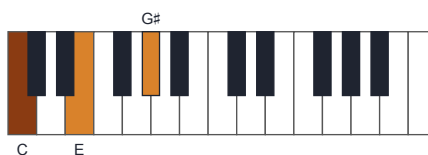
Augmented **Caug**

Formula 1 3 $\sharp$ 5

On C C E G $\sharp$

Also written C+

An unstable, symmetrical triad of stacked major 3rds (major 3rd plus sharp 5th); dreamlike.



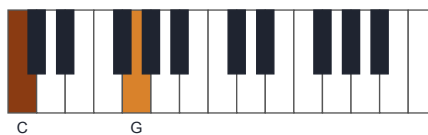
Power Chord **C5**

Formula 1 5

On C C G

Also written called power

Just the root and 5th, no 3rd; neither major nor minor, the staple of rock and metal.

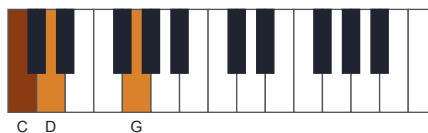


Suspended 2 **Csus2**

Formula 1 2 5

On C C D G

A suspended triad replacing the 3rd with the 2nd; open and unresolved.



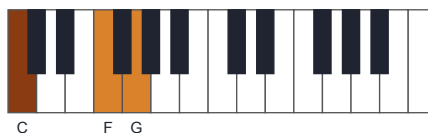
Suspended 4 **Csus4**

Formula 1 4 5

On C C F G

Also written Csus

A suspended triad replacing the 3rd with the 4th; tense, wanting to fall to the major 3rd.



Sixth and seventh chords

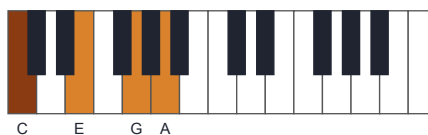
Major 6 **C6**

Formula 1 3 5 6

On C C E G A

Also written Cmaj6, CM6

A major triad plus the 6th; sweet and nostalgic, common in jazz and swing.



Minor 6 **Cm6**

Formula 1 $\flat$ 3 5 6

On C C E $\flat$ G A

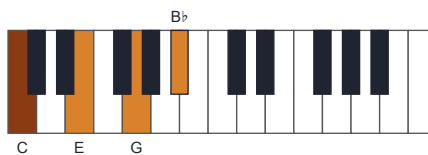
Also written Cmin6

A minor triad plus a major 6th; bittersweet, a favorite in jazz and bossa nova.

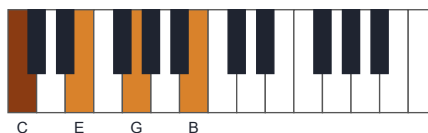


Dominant 7 C7**Formula** 1 3 5 $\flat 7$ **On** C C E G $B\flat$ **Also written** Cdom7

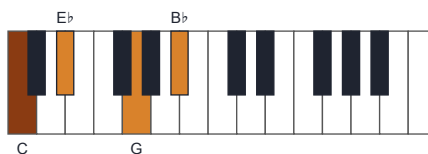
A major triad plus a flat 7th; restless and bluesy, the engine of tension and release.

**Major 7 Cmaj7****Formula** 1 3 5 7**On** C C E G B**Also written** CM7, Cma7, CΔ, CΔ7

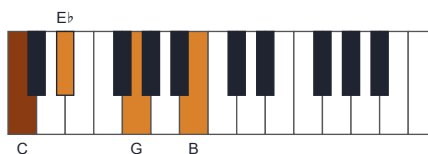
A major triad plus a major 7th; lush, warm and dreamy.

**Minor 7 Cm7****Formula** 1 $\flat 3$ 5 $\flat 7$ **On** C C $E\flat$ G $B\flat$ **Also written** Cmin7, C-7

A minor triad plus a flat 7th; smooth and mellow, the workhorse of jazz and soul.

**Minor Major 7 CmMaj7****Formula** 1 $\flat 3$ 5 7**On** C C $E\flat$ G B**Also written** CmM7, Cminmaj7, Cm(maj7), CmΔ7, CmΔ

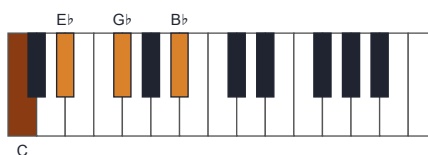
A minor triad with a major 7th; haunting and noir, famous from spy and horror themes.

**Diminished 7 Cdim7****Formula** 1 $\flat 3$ $\flat 5$ $\flat \flat 7$ **On** C C $E\flat$ $G\flat$ $B\flat\flat$ **Also written** C°7, Cø7

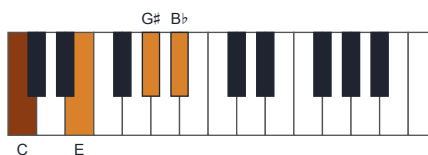
Four stacked minor 3rds; fully symmetrical, maximally tense and endlessly modulating.

**Half-diminished Cm7 $\flat 5$** **Formula** 1 $\flat 3$ $\flat 5$ $\flat 7$ **On** C C $E\flat$ $G\flat$ $B\flat$ **Also written** Cø, Cø7, Cmin7 $\flat 5$, Cm7($\flat 5$); called half-diminished 7

A diminished triad plus a flat 7th; the classic ii of a minor key, darker than m7 but softer than dim7.

**Augmented 7 C7#5****Formula** 1 3 $\sharp 5$ $\flat 7$ **On** C C E $G\sharp$ $B\flat$ **Also written** Caug7, C+7, C7+, C7aug

A dominant 7th with a raised 5th; an altered dominant that pulls hard toward resolution.

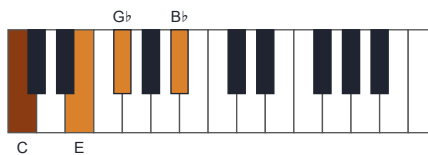


Augmented Major 7 Cmaj7#5**Formula** 1 3 #5 7**On** C C E G# B**Also written** Caugmaj7, C+M7

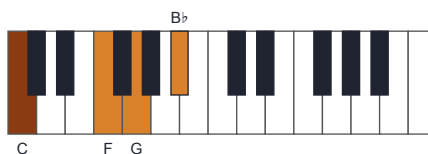
A major 7th with a raised 5th; shimmering and unstable.

**Dominant 7 Flat 5 C7b5****Formula** 1 3 b5 b7**On** C C E Gb Bb

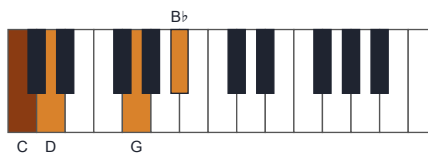
A dominant 7th with a lowered 5th; whole-tone flavored and tense.

**Dominant 7 Sus 4 C7sus4****Formula** 1 4 5 b7**On** C C F G Bb**Also written** C7sus

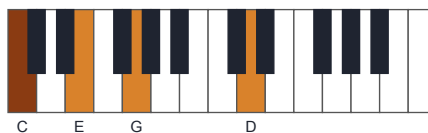
A dominant 7th with a suspended 4th instead of the 3rd; a soulful, delayed resolution.

**Dominant 7 Sus 2 C7sus2****Formula** 1 2 5 b7**On** C C D G Bb

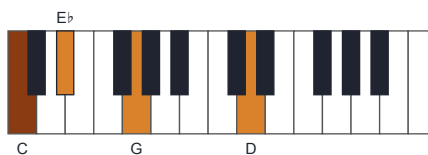
A dominant 7th with the 2nd in place of the 3rd; airy and open.

**Added-tone chords****Add 9 Cadd9****Formula** 1 3 5 9**On** C C E G D**Also written** Cmajadd9

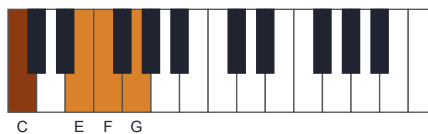
A major triad with an added 9th and no 7th; bright and colorful.

**Minor Add 9 Cmadd9****Formula** 1 b3 5 9**On** C C Eb G D**Also written** Cm(add9)

A minor triad with an added 9th; wistful and modern.

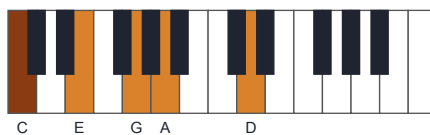
**Add 4 Cadd4****Formula** 1 3 4 5**On** C C E F G**Also written** Cadd11

A major triad with an added 4th; lush and gently clashing, unlike sus4 which omits the 3rd.

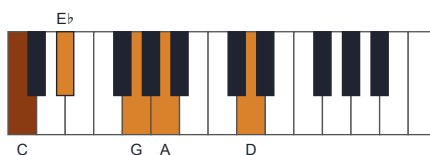


Six Nine C6/9**Formula** 1 3 5 6 9**On** C C E G A D**Also written** C69, C6add9, Cmaj6/9, Cmaj69

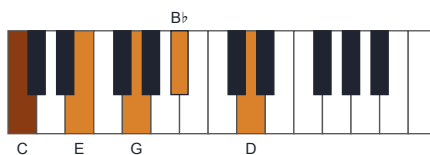
A major 6th chord with an added 9th; a rich, stable jazz ending chord.

**Minor Six Nine Cm6/9****Formula** 1 b3 5 6 9**On** C C E $\flat$ G A D**Also written** Cm69

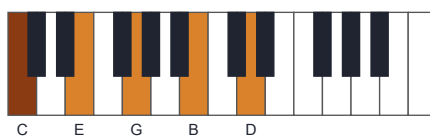
A minor 6th with an added 9th; smooth and sophisticated.

**Ninth chords****Dominant 9 C9****Formula** 1 3 5 b7 9**On** C C E G B $\flat$ D

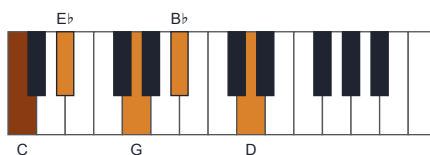
A dominant 7th extended with the 9th; funky and full.

**Major 9 Cmaj9****Formula** 1 3 5 7 9**On** C C E G B D**Also written** CM9

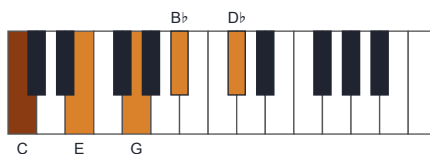
A major 7th extended with the 9th; rich, open and lush.

**Minor 9 Cm9****Formula** 1 b3 5 b7 9**On** C C E $\flat$ G B $\flat$ D**Also written** Cmin9

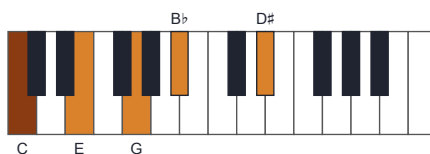
A minor 7th extended with the 9th; warm and soulful.

**Dominant 7 Flat 9 C7b9****Formula** 1 3 5 b7 b9**On** C C E G B $\flat$ D $\flat$

A dominant 7th with a flat 9th; dark and tense, resolving strongly to minor.

**Dominant 7 Sharp 9 C7#9****Formula** 1 3 5 b7 #9**On** C C E G B $\flat$ D $\sharp$ **Also written** called hendrix

A dominant 7th with a sharp 9th; the gritty 'Hendrix chord' clashing major and minor 3rds.



Sharp-eleven (lydian) chords

Dominant 7 Sharp 11 **C7#11**

Formula 1 3 5 $\flat 7$ $\sharp 11$

On C C E G $B\flat$ $F\sharp$

A dominant 7th with a raised 11th; lydian-dominant color, bright and edgy.



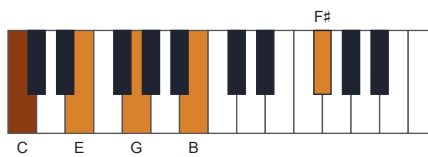
Major 7 Sharp 11 **Cmaj7#11**

Formula 1 3 5 7 $\sharp 11$

On C C E G B $F\sharp$

Also written Cmaj7#4; called lydian chord

A major 7th with a raised 11th; the floating, modern 'lydian' chord.



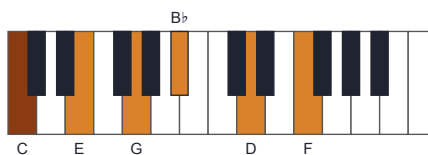
Eleventh and thirteenth chords

Dominant 11 **C11**

Formula 1 3 5 $\flat 7$ 9 11

On C C E G $B\flat$ D F

A dominant 9th extended with the 11th; usually voiced without the 3rd, giving a suspended feel.



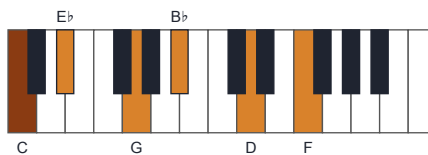
Minor 11 **Cm11**

Formula 1 $\flat 3$ 5 $\flat 7$ 9 11

On C C $E\flat$ G $B\flat$ D F

Also written Cmin11

A minor 9th extended with the 11th; spacious and modal, a quintessential modern-jazz minor sound.

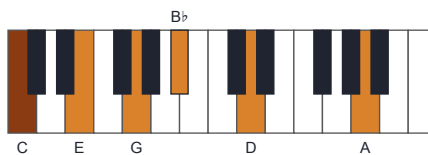


Dominant 13 **C13**

Formula 1 3 5 $\flat 7$ 9 13

On C C E G $B\flat$ D A

A dominant chord extended to the 13th; full and brassy, the lush dominant of big-band jazz.



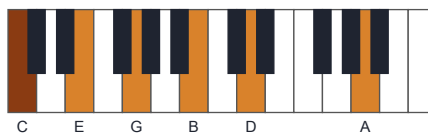
Major 13 **Cmaj13**

Formula 1 3 5 7 9 13

On C C E G B D A

Also written CM13

A major 7th extended to the 13th; maximally lush and resolved.



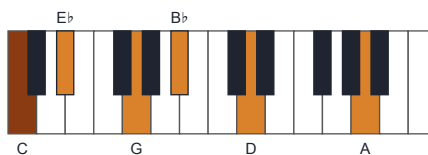
Minor 13 **Cm13**

Formula 1 $\flat 3$ 5 $\flat 7$ 9 13

On C C $E\flat$ G $B\flat$ D A

Also written Cmin13

A minor 7th extended to the 13th; deep, warm and modal.



CHAPTER 5

Harmony in a key

Build a triad on every degree of a scale and you have the chords of that key — the raw material of every progression.

Stacking thirds from each degree of C major, using only notes of the scale, gives seven chords. Their qualities follow from the scale's step pattern, and they are the same in every major key: I ii iii IV V vi vii°. Uppercase numerals are major chords, lowercase minor, ° diminished.

C major — triads

Degree	Numeral	Chord	Notes	Function
1 · tonic	I	C	C E G	tonic
2 · supertonic	ii	Dm	D F A	subdominant
3 · mediant	iii	Em	E G B	tonic
4 · subdominant	IV	F	F A C	subdominant
5 · dominant	V	G	G B D	dominant
6 · submediant	vi	Am	A C E	tonic
7 · leading tone	vii°	Bdim	B D F	dominant

C major — seventh chords

Degree	Numeral	Chord	Notes	Function
1 · tonic	IΔ7	Cmaj7	C E G B	tonic
2 · supertonic	ii7	Dm7	D F A C	subdominant
3 · mediant	iii7	Em7	E G B D	tonic
4 · subdominant	IVΔ7	Fmaj7	F A C E	subdominant
5 · dominant	V7	G7	G B D F	dominant
6 · submediant	vi7	Am7	A C E G	tonic
7 · leading tone	viiø7	Bm7b5	B D F A	dominant

Harmonic function

Chords group into three families by what they do. **Tonic** chords (I, and its substitutes vi and iii) feel like home. **Subdominant** chords (IV, ii) move away. **Dominant** chords (V, vii°) create tension that wants to resolve home: their leading tone (B in C major) pulls up to the tonic and their 7th (F) pulls down to the 3rd. Most progressions cycle tonic → subdominant → dominant → tonic.

Minor keys

Natural minor lacks a leading tone, so its v chord is minor and weak. Composers borrow the raised 7th of harmonic minor to get a major V (E–G#–B in A minor) and the powerful

V7. Numerals in this book are written relative to the major scale: in A natural minor the chord on the third degree is $\flat$ III.

A natural minor — triads

Degree	Numeral	Chord	Notes	Function
1 · tonic	i	Am	A C E	tonic
2 · supertonic	ii $^{\circ}$	Bdim	B D F	subdominant
3 · mediant	$\flat$ III	C	C E G	tonic
4 · subdominant	iv	Dm	D F A	subdominant
5 · dominant	v	Em	E G B	dominant
6 · submediant	$\flat$ VI	F	F A C	tonic
7 · subtonic	$\flat$ VII	G	G B D	dominant

A harmonic minor — seventh chords

Degree	Numeral	Chord	Notes	Function
1 · tonic	i Δ 7	AmMaj7	A C E G $\sharp$	tonic
2 · supertonic	ii $\emptyset$ 7	Bm7 $\flat$ 5	B D F A	subdominant
3 · mediant	$\flat$ III+ Δ 7	Cmaj7 $\sharp$ 5	C E G $\sharp$ B	tonic
4 · subdominant	iv7	Dm7	D F A C	subdominant
5 · dominant	V7	E7	E G $\sharp$ B D	dominant
6 · submediant	$\flat$ VI Δ 7	Fmaj7	F A C E	tonic
7 · leading tone	vii $^{\circ}$ 7	G $\sharp$ dim7	G $\sharp$ B D F	dominant

Is a chord in the key?

A chord belongs to a key only if **every** one of its notes is in the scale — not just its root. E7 in A natural minor stands on degree 5, but its G $\sharp$ is foreign, so it is a *chromatic* (borrowed) chord: the analysis gives V7 with non-scale note G $\sharp$. In A harmonic minor the same E7 is fully diatonic.

Analyzing C – A7 – Dm – G7 – C in C major:

Chord	Numeral	In key?	Why
C	I	yes	tonic
A7	VI7	no	contains C $\sharp$
Dm	ii	yes	subdominant
G7	V7	yes	dominant
C	I	yes	tonic

Pro note. Numerals follow spelling: in C, a G $\flat$ chord is $\flat$ V and an F $\sharp$ chord $\sharp$ IV, even though they sound the same. The A7 above is chromatic because of its C $\sharp$; functionally it is the dominant of ii (V7/ii), which chapter 7 builds on purpose.

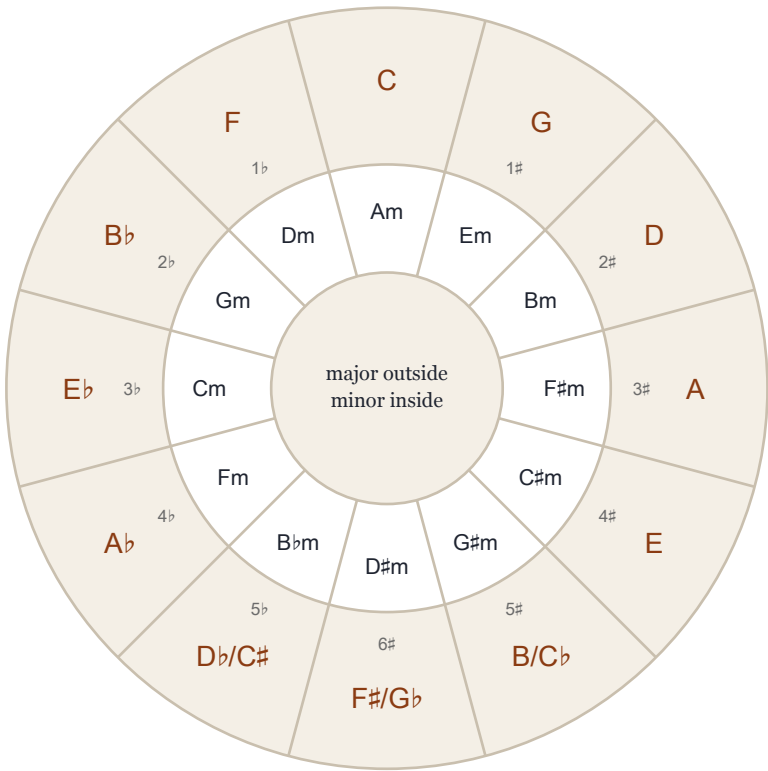
Common progressions

Name	Numerals	In C major
Pop axis	I V vi IV	C G Am F
Doo-wop	I vi IV V	C Am F G
Jazz ii–V–I	ii V I	Dm G C
Circle descent	vi ii V I	Am Dm G C
Royal road	IV V iii vi	F G Em Am
Plagal cadence	IV I	F C
Andalusian cadence	i ♭VII ♭VI V	Am G F E (in A minor; V from harmonic minor)

CHAPTER 6

The circle of fifths

Arrange the twelve keys a fifth apart and they form a circle in which neighbours share all but one note.



Moving clockwise raises the key by a perfect fifth and adds one sharp (or removes a flat); moving counter-clockwise lowers it by a fifth and adds a flat. The inner ring shows each major key's **relative minor**, which shares its key signature.

Major	Signature	Accidentals	Relative minor
C	none	—	A minor
G	1 sharp	F#	E minor
D	2 sharps	F# C#	B minor
A	3 sharps	F# C# G#	F# minor
E	4 sharps	F# C# G# D#	C# minor
B / Cb	5 sharps	F# C# G# D# A#	G# minor
F# / Gb	6 sharps	F# C# G# D# A# E#	D# minor
Db / C#	5 flats	Bb Eb Ab Db Gb	Bb minor
Ab	4 flats	Bb Eb Ab Db	F minor
Eb	3 flats	Bb Eb Ab	C minor

Major	Signature	Accidentals	Relative minor
B♭	2 flats	B♭ E♭	G minor
F	1 flat	B♭	D minor

Closely related keys

The keys one step either side, plus the relative minors of all three, differ by at most one accidental. From C major they are A minor, G major, E minor, F major and D minor. They are the natural destinations for a modulation or a contrasting bridge: the change is audible but smooth.

Sharps are added in the order F C G D A E B, flats in the reverse order B E A D G C F. A key with more than seven accidentals is *theoretical* — D# major would need double sharps — and is written as its enharmonic instead (E♭ major).

CHAPTER 7

Voice leading and reharmonization

How chords move matters as much as which chords you choose. Then: three classic ways to recolour a progression.

Play every chord in root position and the whole hand jumps. Good **voice leading** treats each chord note as a singer: keep notes that two chords share (**common tones**) where they are, and move the others to the nearest available note. The best voicing is the one with the least total movement, each voice going to exactly one voice of the next chord.

Chord	Voicing (low → high)	Movement
C	C ₄ E ₄ G ₄	—
Am	C ₄ E ₄ A ₄	2 semitones
F	C ₄ F ₄ A ₄	1 semitone
G	D ₄ G ₄ B ₄	6 semitones
C	E ₄ G ₄ C ₅	3 semitones

A slash chord keeps its named bass at the bottom and voice-leads the rest above it.

Secondary dominants

Any major or minor chord of a key can be made to feel like a temporary tonic by preceding it with *its own* dominant — the dominant-seventh chord a perfect fifth above it. This is the **secondary dominant**, written V₇/x.

Target	Its dominant	Notes	Label
Dm	A ₇	A C [♯] E G	V ₇ /ii
Em	B ₇	B D [♯] F [♯] A	V ₇ /iii
F	C ₇	C E G B [♭]	V ₇ /IV
G	D ₇	D F [♯] A C	V ₇ /V
Am	E ₇	E G [♯] B D	V ₇ /vi

Tritone substitution

A dominant seventh's two guide tones — its 3rd and 7th — form a tritone. The dominant a tritone away contains the *same* tritone, spelled the other way round (G₇ has B and F; D[♭]₇ has F and C[♭]). So D[♭]₇ can stand in for G₇, and the bass slides down by semitone into C. Only dominants can be substituted this way; a plain major triad is read as the dominant seventh it implies.

Dominant	Substitute	Shared guide tones
G ₇	D [♭] ₇	B and F
C ₇	G [♭] ₇	E and B [♭]

Dominant	Substitute	Shared guide tones
D7	A $\flat$ 7	F $\sharp$ and C
E7	B $\flat$ 7	G $\sharp$ and D
A7	E $\flat$ 7	C $\sharp$ and G
B7	F7	D $\sharp$ and A

Negative harmony

Negative harmony (after Ernst Levy, popularized by Jacob Collier) mirrors every note around the axis halfway between the tonic and its fifth. In C the axis lies between E and E $\flat$, so E $\leftrightarrow$ E $\flat$, G $\leftrightarrow$ C, D $\leftrightarrow$ F, and a major chord becomes a minor one: the tonic C major turns into C minor, the dominant G into F minor. Functions survive — the mirror of a dominant still pulls home — but the colour darkens. The rule is the same in every key: a note r semitones above the tonic maps to $7 - r$ semitones above it.

Note	C	C $\sharp$	D	E $\flat$	E	F	F $\sharp$	G	A $\flat$	A	B $\flat$	B
Mirror in C	G	G $\flat$	F	E	E $\flat$	D	D $\flat$	C	B	B $\flat$	A	A $\flat$

Chord in C	Mirror	Chord in D	Mirror
C E G	G E $\flat$ C	D F $\sharp$ A	A F D
F A C	D B $\flat$ G	G B D	E C A
G B D	C A $\flat$ F	A C $\sharp$ E	D B $\flat$ G
A C E	B $\flat$ G E $\flat$	B D F $\sharp$	C A F

CHAPTER 8

Melody craft

Melodies are built from scale steps, shaped by repetition and variation, and fitted to the harmony.

Scale degrees and the number line

Writing a melody as scale degrees — 1 2 3 5 8 — makes it portable: the same numbers give a melody in any key or mode. Degrees above 7 continue into the next octave (8 is the tonic an octave up, 9 the 2nd). Below the tonic the line continues without a zero: -1 is the step just below the tonic, -2 the step below that, -7 the tonic an octave down.

Degrees	-3	-1	1	2	3	5	8
From C4, major	G3	B3	C4	D4	E4	G4	C5
From A3, natural minor	E3	G3	A3	B3	C4	E4	A4

Motifs: repetition and variation

Memorable melodies repeat. A **motif** — a short cell — is stated, repeated, and varied. The classic variations are **transposition** (shift every note by the same number of scale steps), **inversion** (mirror the contour: up becomes down), **retrograde** (play it backwards) and **rotation** (start it from a different note). Forms such as AABA or ABAC arrange the results.

Label	Rule	Degrees	In C major
A	the motif	1 2 3 5	C5 D5 E5 G5
B	A transposed down 2 steps	-2 -1 1 3	A4 B4 C5 E5
C	A inverted	1 -1 -2 -4	C5 B4 A4 F4

Sequences

A **sequence** restates a motif at successive pitch levels, usually a step apart — the engine of Baroque episodes and countless pop bridges. A *diatonic* sequence stays in the key, so the interval qualities change slightly from copy to copy; a chromatic passing note keeps its distance from the scale note beneath it.

Harmonizing a melody

To choose chords under a melody, list the chords that contain each melody note, then prefer the one that shares the most notes with the previous chord. That single rule — maximum common tones — produces smooth, singable harmony; ties go to the melody note sitting low in the chord (the root first), then to familiar chord qualities, then to simpler chords.

Pro note. Two refinements make the rule musical: *snap to scale* (move stray notes to the nearest scale note, ties downward) keeps a line compatible with diatonic chords, and forbidding the same pitch set twice in a row (C6 after Am7 is the same four notes) forces genuine harmonic motion.

Tintinnabuli (Arvo Pärt)

Pärt's *tintinnabuli* technique pairs a stepwise melodic voice (the M-voice) with a voice that only ever plays notes of one triad (the T-voice) — like a bell ringing the tonic chord. For each melody note the T-voice takes the nearest triad note above it (*superior*), below it (*inferior*), or alternately; 'second position' skips to the next triad note. The T-voice can never clash with the harmony, which is why the texture sounds so luminous.

M-voice	A4	B4	C5	D5	C5	B4	A4
T-voice above	C5	C5	E5	E5	E5	C5	C5
T-voice below	E4	A4	A4	C5	A4	A4	E4

CHAPTER 9

Species counterpoint

Johann Joseph Fux's 1725 method for writing independent melodies that sound good together — still the best course in voice independence.

Counterpoint adds a new melody to a given one, the **cantus firmus** (a plain line in whole notes, beginning and ending on the tonic). Fux taught it in five **species** of increasing rhythmic freedom. The rules below are the ones the counterpoint tool enforces; intervals are judged by their spelling, so a diminished seventh is a dissonance even though it spans as many semitones as a major sixth.

Rules for every species

- **Strong beats are consonant:** unison/octave, fifth, thirds and sixths. The perfect fourth counts as a dissonance in two voices.
- **Begin and end on a perfect consonance;** the last note is the tonic, reached by step (usually the leading tone rising, or the 2nd falling). A lower voice begins on the unison or octave — a fifth below would suggest another key.
- **No parallel or direct fifths and octaves:** never move both voices in the same direction into a perfect consonance. Contrary motion is best, oblique next, similar motion into imperfect consonances acceptable.
- **Melodic lines sing:** no leaps of a tritone, a seventh, or any augmented or diminished interval, and none larger than an octave. Steps are the norm; leaps are rare and small.
- **Keep the voices apart but close:** the counterpoint stays on its side of the cantus, generally within a tenth.

The five species

- **First species** (1:1): note against note, all consonant; imperfect consonances preferred in the middle.
- **Second species** (2:1): two notes per cantus note. The first is consonant; the second may be a dissonant *passing tone* that fills a third by step in one direction. No note is struck twice in a row inside a bar.
- **Third species** (4:1): four notes per cantus note. Dissonances are passing or neighbour tones, approached and left by step, and never next to another dissonance.
- **Fourth species** (syncopation): each note is struck on the weak beat as a consonance and tied over the barline. If the tied note now clashes, it is a *suspension* and must resolve down by step to a consonance: 7–6, 4–3 or 9–8 above the cantus; 2–3, 4–5 or 9–10 below. A 7–8 resolution below is forbidden, and parallels are judged between the struck weak-beat notes.
- **Fifth species** (florid): a free mix of the others, with a prepared suspension at the cadence.

One cantus, five species

The cantus C5 D5 E5 F5 E5 D5 C5 in C major with a counterpoint above it. The table shows the counterpoint and the interval on each downbeat.

Species	Counterpoint	Downbeat intervals
1 (1:1)	C6 B5 G5 A5 G5 B5 C6	P1/P8 M6 m3 M3 m3 M6 P1/P8
2 (2:1)	C6 G5 F5 A5 G5 B5 A5 B5 C6 G5 A5 B5 C6	P1/P8 m3 m3 M3 m6 P5 P1/P8
3 (4:1)	C6 B5 A5 G5 F5 G5 A5 D6 C6 B5 A5 B5 A5 G5 A5 B5 C6 B5 A5 G5 A5 G5 A5 B5 C6	P1/P8 m3 m6 M3 m6 P5 P1/P8
4 (syncopated)	C6 E6 D6 C6 D6 C6 B5 C6	P1/P8 M2 m7 P5 m7 m7 P1/P8
5 (florid)	C6 B5 A5 G5 F5 A5 G5 A5 B5 E6 D6 C6 G6 C6 B5 C6	P1/P8 m3 m3 M6 m3 m7 P1/P8

Why these rules. Parallel fifths and octaves are forbidden because perfect consonances blend so completely that two voices moving in parallel fuse into one — the independence the exercise is about disappears. Dissonance handling (prepare, sound on a weak beat or as a suspension, resolve by step) is the grammar of tension and release that tonal music still runs on.

CHAPTER 10

Rhythm and form

Patterns in time: onset grids, Euclidean rhythms, the grooves of world music, and the large-scale shape of a song.

Rhythm as a grid

A rhythm can be written as a row of steps: **O** a strong (accented) hit, **o** a weak hit, **.** a rest. Sixteen steps of a quarter beat each make one bar of 4/4 in sixteenth notes; eight steps of half a beat make a bar in eighths.

Euclidean rhythms

Spread *k* hits as evenly as possible over *n* steps (Bjorklund's algorithm, as described by Godfried Toussaint) and a remarkable number of traditional rhythms appear. Three hits in eight steps is the Cuban *tresillo*; five in eight the *cinquillo*; five in sixteen the bossa-nova bass.

Hits / steps	Pattern	Known as
3 / 8	O..o.o.	tresillo (Cuba)
5 / 8	O.oo.oo.	cinquillo (Cuba)
2 / 5	O.o..	khafif-e-ramal (13th-century Persia)
3 / 7	O.o.o..	ruchenitza (Bulgaria)
4 / 9	O.o.o.o..	aksak (Turkey)
5 / 12	O..o.o..o.o.	Venda clapping (South Africa)
7 / 12	O.oo.o.oo.o.	the standard West African bell pattern
5 / 16	O..o.o..o..o...	bossa-nova bass (Brazil)
7 / 16	O..o.o.o..o.o.o.	a samba necklace (Brazil)

A groove library

Groove	Pattern	Description
four on floor	O...O...O...O...	A kick on every beat — house/techno/disco pulse.
backbeat	O.....O...	Snare on beats 2 and 4 — the backbone of rock and pop.
offbeat	..o...o...o...o.	Hits on the off-beats (the 'and's) — house open hats, ska/reggae skank.
eighths	o.o.o.o.o.o.o.o.	Steady eighth notes — driving hats or a running pulse.
sixteenths	oooooooooooooooo	Steady sixteenths — rolling hats or trance arps.
tresillo	O..o.o.	The 3+3+2 Cuban/Latin cell, ubiquitous in pop and reggaeton.
cinquillo	O.oo.oo.	A five-stroke Cuban cell, denser sibling of the tresillo.

Groove	Pattern	Description
habanera	O..oo.o.	The habanera/tango rhythm — dotted then even.
son clave 32	O..o..o...o.o...	3-2 son clave, the key pattern of Afro-Cuban son and salsa.
rumba clave 32	O..o...o..o.o...	3-2 rumba clave — like son clave but the 'three' side is shifted.
bossa nova	O..o..o...o..o..	A bossa-nova clave variant — gentle Brazilian sway.
dembow	O..oO.o.O..oO.o.	The reggaeton 'dembow' — boom-ch-boom-chick engine.

Song form

A song is a sequence of sections. Letter forms name repeated material: AABA is the 32-bar standard (three statements of a strain around a contrasting bridge); verse–chorus forms alternate a storytelling verse with a recurring chorus, often with a pre-chorus to build into it and a bridge for contrast before the last chorus. Conventional lengths, in bars:

Section	Bars	Role
intro	4	sets the mood, often a reduced groove
verse	8	tells the story; lower energy
prechorus	4	builds tension into the chorus
chorus	8	the hook: highest energy, repeated words
bridge	8	contrast — new chords, often a new key area
outro	4	winds down or fades
drop	8	the energy release in dance music
break	4	a pause: instruments drop out
fill	1	a one-bar transition, usually drums
hook	8	a short, repeated signature phrase

Try it. Take one four-chord loop, a motif and a groove. Write the verse with the motif low and sparse, the chorus with the motif transposed up and the full groove, and a bridge on the subdominant key from the circle of fifths. That is a complete, coherent song built entirely from the rules in this book.

CHAPTER 11

Rules of the old masters

Melody was codified long before harmony: chant modes, Guido's syllables, the cantus firmus. Then Rameau, Bach and Schoenberg put harmony into rules of their own.

The eight church modes

Gregorian chant is organised in eight modes, four pairs sharing a **final** (the note a melody ends on): D, E, F and G. Each *authentic* mode spans the octave above its final; its *plagal* partner (hypo-) lies a fourth lower. The **tenor** or reciting tone is the pitch psalms are chanted on — the melodic centre of gravity.

Mode	Name	Final	Tenor	Ambitus
I	Dorian	D	A	D4 – D5
II	Hypodorian	D	F	A3 – A4
III	Phrygian	E	C	E4 – E5
IV	Hypophrygian	E	A	B3 – B4
V	Lydian	F	C	F4 – F5
VI	Hypolydian	F	A	C4 – C5
VII	Mixolydian	G	D	G4 – G5
VIII	Hypomixolydian	G	C	D4 – D5

Guido's hexachords

Around 1030 Guido of Arezzo taught singers six syllables — **ut re mi fa sol la** — from the hymn *Ut queant laxis*. They name a hexachord, not a key: *naturale* on C, *durum* on G (with B natural) and *molle* on F (with B $\flat$). The semitone is always **mi–fa**. To go beyond six notes a singer **mutates** to another hexachord on a pivot note that belongs to both — by the old rule *per re sursum, per la deorsum*: rising, the pivot is renamed re; falling, la. Notes outside the gamut (F $\sharp$, E $\flat$) are *musica ficta*, sung in a transposed hexachord: D E F $\sharp$ G is ut re mi fa.

Note	C	D	E	F	G	A	B	C
Syllable	ut	re	mi	fa	sol	re	mi	fa

The mutation happens on A: la (*naturale*) becomes re (*durum*).

Guido's vowel method

In the *Micrologus* (c. 1026) Guido proposed a way to find a melody for any text: write the vowels a e i o u under the notes of the scale, starting on gamma (the low G) and repeating, then sing each syllable on a note that carries its vowel. It is the oldest written recipe for composing by rule. (The table below sits in the modern octave of the modes, so the Dorian

final D carries u, as in Guido's own table; a second row starting on B widens the choice.)

Vowel	u	e	a	a	i	e	o	a	e	i	i	
Note	D4	F4	E4	E4	G4	F4	A4	C5	D5	G4	G4	D4

Here each vowel takes the candidate nearest the previous note inside the Dorian range, and the tune closes on the final D.

The cantus firmus

Fux's pupils first learned to write a plain, singable line. The rules below are the ones modern species-counterpoint teaching distils from Fux and Jeppesen (Fux's own cantus firmi occasionally break the leap rules; the leading-tone rule comes from tonal teaching):

- Begin and end on the final; reach the last note by step.
- Move mostly by step; allowed leaps are thirds, fourths, fifths, the ascending minor sixth and the octave — never a tritone, a seventh or an augmented/diminished interval.
- After a leap larger than a third, turn back by step (a gap wants filling).
- No more than two leaps in a row; two leaps in the same direction must outline a consonance.
- One climax only, somewhere in the middle; a range of about an octave, never beyond a tenth.
- No repeated notes, no restated figures, no long scale runs.
- In the Dorian, Mixolydian and Aeolian modes the note before the final may be raised a semitone (*musica ficta*), as singers did at a cadence.

A cantus firmus written by these rules in D Dorian: D4 A3 B3 A3 D4 C4 D4 B3 C4 E4 D4.

Rameau: the fundamental bass

Jean-Philippe Rameau (*Traité de l'harmonie*, 1722) argued that every chord grows from a root, whatever note lies in the bass, and that good harmony moves its **fundamental bass** by fifths, then thirds. He named the cadences by that motion: *cadence parfaite* when the fundamental bass falls a fifth from dominant to tonic, *cadence irrégulière* when it rises a fifth (IV–I or I–V), *cadence rompue* for V–vi.

Under C F/A G7 C Am Dm G C the fundamental bass is C F G C A D G C: 5 moves by fifth, 1 by third, 1 by step.

Schoenberg: ascending, descending and superstrong progressions

In *Structural Functions of Harmony* (1954) Arnold Schoenberg classified root motion by what it does to the previous root. **Ascending** (strong) progressions — root up a fourth or down a third — absorb the old root as a lesser member of the new chord and push forward. **Descending** progressions — root up a fifth or up a third — do the opposite (later writers

call them weak; Schoenberg avoided the word). **Superstrong** progressions move the root by step and share no tones. He advised building mainly on ascending ones.

For the same progression: ascending (strong): 6, superstrong: 1.

Bach: four-part chorale writing

J. S. Bach's 371 four-part chorales became the textbook of harmony; the rules below are the ones teachers distilled from them. Four voices (soprano, alto, tenor, bass) in their ranges; upper voices within an octave of each other; the bass on the root or the given inversion; complete chords, root doubled by preference, never the leading tone; no parallel fifths or octaves between any two voices; no voice overlapping its neighbour; the leading tone rises and a chord seventh falls.

Voice	C	Am	F	G7	C
soprano	G4	A4	A4	G4	G4
alto	C4	C4	C4	B3	C4
tenor	E3	E3	F3	F3	E3
bass	C3	A2	F2	G2	C3

CHAPTER 12

Modern approaches

Twentieth-century composers and theorists found new rules: transformations, axes, rows, sets and processes.

Neo-Riemannian transformations

Following Hugo Riemann, David Lewin and Richard Cohn describe three moves between major and minor triads that keep two notes and move the third by a step: **P** (parallel: C ↔ Cm), **R** (relative: C ↔ Am) and **L** (leading-tone exchange: C ↔ Em). Chaining them produces the smooth, key-less triad progressions heard in late Romantic and film music.

C, then L R L R: C → Em → G → Bm → D.

Bartók's axis system

Ernő Lendvai showed that Bartók treats keys a minor third apart as functional equivalents. Each function forms an axis of four keys — a pole, its counterpole a tritone away, and two secondary poles.

Axis	Pole	Counterpole	Secondary poles
tonic	C	F#	A Eb
subdominant	F	B	D Ab
dominant	G	C#	E Bb

Coltrane changes

In *Giant Steps* (1959) John Coltrane replaced the ii–V–I with a whirl through three tonal centres a major third apart, each approached by its own dominant.

Original	Dm7	G7	Cmaj7
Coltrane	Dm7	Eb7	Abmaj7 B7 Emaj7 G7 Cmaj7

Schoenberg's twelve-tone method

A **row** orders all twelve pitch classes; the piece is built from its 48 forms — the prime (P), inversion (I), retrograde (R) and retrograde inversion (RI), each in twelve transpositions. The matrix shows them all: rows read left to right are P forms, columns top to bottom are I forms.

	I4	I5	I7	I1	I6	I3	I8	I2	I11	I10	I9	I10
P4	E	F	G	Db	Gb	Eb	Ab	D	B	C	A	Bb
P3	Eb	E	Gb	C	F	D	G	Db	Bb	B	Ab	A
P1	Db	D	E	Bb	Eb	C	F	B	Ab	A	Gb	G

	I4	I5	I7	I1	I6	I3	I8	I2	I11	I10	I9	I10
P7	G	A♭	B♭	E	A	G♭	B	F	D	E♭	C	D♭
P2	D	E♭	F	B	E	D♭	G♭	C	A	B♭	G	A♭
P5	F	G♭	A♭	D	G	E	A	E♭	C	D♭	B♭	B
P0	C	D♭	E♭	A	D	B	E	B♭	G	A♭	F	G♭
P6	G♭	G	A	E♭	A♭	F	B♭	E	D♭	D	B	C
P9	A	B♭	C	G♭	B	A♭	D♭	G	E	F	D	E♭
P8	A♭	A	B	F	B♭	G	C	G♭	E♭	E	D♭	D
P11	B	C	D	A♭	D♭	B♭	E♭	A	G♭	G	E	F
P10	B♭	B	D♭	G	C	A	D	A♭	F	G♭	E♭	E

Forms are labelled by their first pitch class (C = 0); this row is Schoenberg's own, from the Suite for Piano, Op. 25.

Pitch-class sets

Allen Forte and John Rahn classify any collection of pitch classes by its **prime form** — the most compact version of it or of its inversion, transposed to start on 0 — and its **interval vector**, the count of each interval class between its notes. Every major and minor triad shares one set class. (Prime forms here follow Rahn's packing, as most textbooks now do; Forte's original table packs six set classes differently.)

Collection	Prime form	Interval vector
C major triad	[0,3,7]	<001110>
A minor triad	[0,3,7]	<001110>
Diminished seventh	[0,3,6,9]	<004002>
Whole-tone scale	[0,2,4,6,8,10]	<060603>
C major scale	[0,1,3,5,6,8,10]	<254361>

Messiaen's modes of limited transposition

Olivier Messiaen used seven symmetric scales that return to themselves after a few transpositions — the whole-tone scale is mode 1, the half-whole octatonic scale mode 2; modes 3 to 7 appear in the scale catalogue of chapter 3.

Minimalist processes

Philip Glass builds music by **additive process**: a figure grows one note at a time (C4 D4 | C4 D4 E4 | C4 D4 E4 F4). Steve Reich's **phasing** sets a pattern against a copy of itself that slips one step ahead at each stage until the two voices meet again — the process of *Piano Phase* and *Clapping Music*.

APPENDIX

Glossary

Term	Meaning
Accidental	A sign that raises (#) or lowers (b) a note by a semitone; doubled for two.
Cadence	A closing progression, e.g. V–I (authentic), IV–I (plagal), V–vi (deceptive).
Cantus firmus	The given melody against which counterpoint is written.
Chromatic	Using notes outside the key; a chromatic chord contains at least one.
Common tone	A note shared by two consecutive chords.
Degree	A note's position in a scale, 1 for the tonic, written relative to major (♭3, #4).
Diatonic	Belonging to the key: made only of scale notes.
Dominant	The 5th degree and the chord on it; the chord of tension that resolves to the tonic.
Enharmonic	Two spellings of the same pitch, such as F# and G♭.
Guide tones	The 3rd and 7th of a chord, which define its quality.
Interval	The distance between two notes, named by letter count and quality.
Inversion	A chord with a note other than the root in the bass; or an interval turned upside down.
Leading tone	The note a semitone below the tonic, which pulls up to it.
Mode	A scale that starts on another degree of a parent scale.
Parallel motion	Two voices moving in the same direction by the same interval.
Pitch class	All notes with one name regardless of octave: every C.
Relative key	The major and minor keys sharing one key signature (C major and A minor).
Suspension	A consonant note held over into a chord where it clashes, then resolved down by step.
Tonic	The home note of a key or scale.
Tritone	The interval of three whole tones (six semitones): augmented 4th or diminished 5th.
Voice leading	The way individual lines move from chord to chord.

About this book

This book is generated by `docs/book/build_book.py` in the `midi-composer-mcp` repository. The catalogue entries, tables and worked examples are computed by the same deterministic functions the composing tools expose to an AI assistant — the scale and chord databases, the diatonic harmony resolver, the analysis and voice-leading rules, the reharmonization tools, the tintinnabuli and species-counterpoint engines, and the rhythm generators. When a rule in the code changes, rebuilding the book updates every example with it.